END-TERM PROJECT PRESENTATION

SMART TRANSPORTATION NETWORK PLANNING AND RESILIENCE ANALYSIS

SYSTEM MODELING, ALGORITHMIC OPTIMIZATION, & FAULT TOLERANCE

GRAPH SIZE CONSTRAINTS: 500 CITIES MINIMUM

PROBLEM STATEMENT

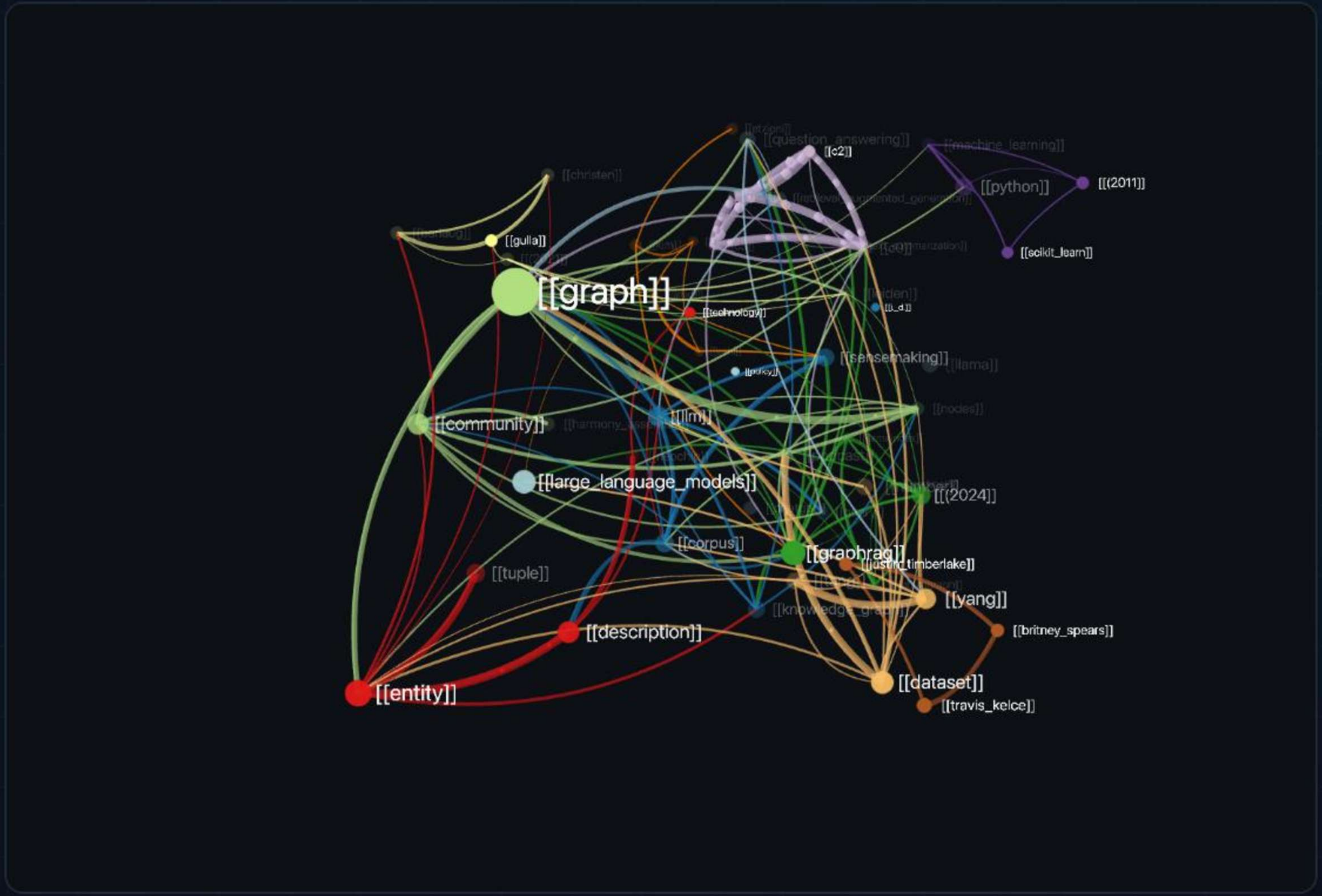
Core Objective

The government plans to develop a smart transportation network connecting 500 cities. The objective is to minimize infrastructure costs, identify strategic cities for future development, and ensure network resilience against disasters and road failures.

Graph Modeling Framework

The network is mathematically modeled as a weighted graph $G = VE$ where:

- **Vertices (V):** Represent the individual cities.
- **Edges (E):** Represent physical roads connecting the cities.
- **Edge Weights:** Represent dynamic construction or travel costs.
- **Scale Constraints:** System spans a minimum of 500 cities. Random graph generation may be utilized.



TASK 1: MST OPTIMIZATION

Minimum Spanning Tree Execution

Constructs a complete, cost-minimized network connecting all 500 nodes with absolute efficiency.

- **Algorithmic Methods:** Implementation of Kruskal's Algorithm or Prim's Algorithm.
- **Target Metric:** Identifies structural connections to minimize total construction costs.
- **Output Specifications:** Displays original network cost, MST network cost, total savings, and exact roads selected.

Learning Outcome
Understand how governments optimize infrastructure investments while maintaining connectivity.

>_ Expected Output Console

```
Original Cost = X
MST Cost = Y
Cost Saved = Z %

Selected Roads:
(1,5)
(5,8)
(8,12)
 ...
[All 499 structural spans successfully written to path buffer]
```

TASK 2: STRATEGIC CITY IDENTIFICATION

Connectivity Centroid Calculations

Identify the most influential nodes using centrality analysis to map urbanization priority indicators.

- **Degree Centrality:** Computed systematically for every vertex.
- **Ranking:** Ranks all nodes by connection counts to print Top 10 cities.
- **Analytical Detail:** Displays City ID, connectivity density, and rank.
- **Strategic Module:** Dynamically triggers automated site suitability recommendations based on computed degrees.

Learning Outcome
Understand how graph centrality helps in urban planning and infrastructure development.

>_ Strategic Output Logs

```
Rank 1 → City 143 → Degree 37
Rank 2 → City 278 → Degree 34
...

Top Development Candidates:
• Airport: City 143
• Logistics Hub: City 278
• Railway Junction: City 96

Recommendations generated:
"City 143 is connected to 37 other cities and is a
strong candidate for an international airport."
```

TASK 3: DISASTER RECOVERY ROUTING

Dynamic Network Disruption Scenarios

Evaluate alternate routes when natural disasters destroy cities (node removal) or roads (edge removal).

- **Step 1:** Allow user to remove: One or more cities OR one or more roads.
- **Step 2:** User inputs Source City A, Source City B, and Destination City D.
- **Step 3:** Compute A → D shortest path and B → D shortest path.

```
Source A Path: Distance = 275 km | Path: 12 → 55 → 91 → 203
Source B Path: Distance = 312 km | Path: 44 → 67 → 103 → 203

Fallback state (If path does not exist):
"No valid route available after disaster."
```

Learning Outcome: Understanding fault-tolerant systems.

TASK 4: TRAFFIC-AWARE SMART ROUTING

Dynamic Edge Weights

Simulates real-world traffic flows by updating the path metrics in real-time based on congestion states.

- **State Categorization:** Each road is marked as either Low Traffic, Medium Traffic, or High Traffic.
- **Dynamic Multipliers:** Edge weights scale up based on real-time traffic levels.
- **System Inputs:** Receives Source City, Destination City, and Traffic Scenario parameters.

Learning Outcome
Real-time route optimization.

>_ Rerouting Output Metrics

```
Normal Route:
Distance = 120 km

Traffic Route:
Distance = 185 km
Delay = 54.1%

[System actively calculated congestion bypass routes
based on elevated edge penalties]
```

TASK 5: CRITICAL INFRASTRUCTURE ANALYSIS

Systemic Vulnerability Analysis

Dianoses specific single points of failure that cause extreme structural fractures inside the system.

- **Node Elimination:** Temporarily remove each city node and measure the reduction in overall network connectivity.
- **Edge Elimination:** Temporarily remove each road edge and measure the relative system disruption.

Learning Outcome
Identification of critical infrastructure.

>_ Critical Diagnostic Output

```
Most Critical City:
City 87

Disconnected Components Created:
12

Most Critical Road:
(87,143)

[Removal of City 87 breaks network graph into 12
disconnected subgrids]
```